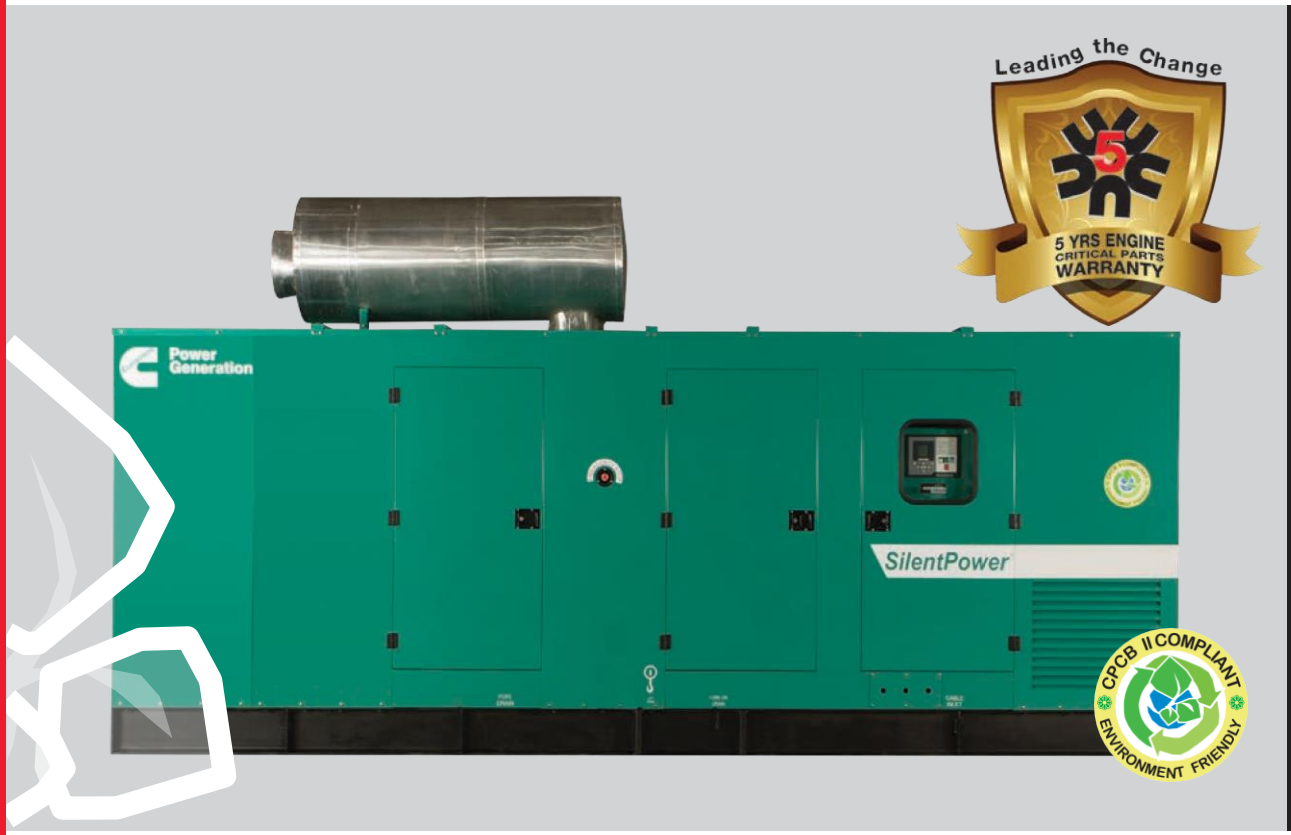
SUDHIR® Rentals

Cummins® Power Generation

Diesel Generator Set

K19 Series

500 kVA, 400 kWe Prime



Specification Sheet

Reliable Technology And Unmatched Performance

- The Cummins® K19 series heavy-duty engine and world class Stamford alternator powered diesel generator set
- Proven technology with mechanical simplicity of Cummins PT fuel system.
- Advanced in-cylinder technology to meet latest emission norms without any after-treatment device.
- Smart aesthetic and superior finish
- Compact in size with optimum power to weight ratio

Environment Friendly Power

- Class defining technology engine is designed to meet stringent exhaust emission tests as per revised MoEF norms, thus offering environment friendly power.
- The Cummins® diesel generator sets are available with the lowest noise levels in its range

Lowest Operating Cost And Comprehensive Warranty

- Highly reliable and durable Product
- All elements are designed to work together to maximize efficiency even at part loads, offering the advantage of lowest operating costs.
- 300 Hours/1 year service interval
- Industry acknowledged best-in-class comprehensive warranty on the entire package including rubber components

Single Source Power Assurance

- All the major components – the engine, alternator, control system and canopy are designed, manufactured and tested by Cummins India.
- Best and largest customer support network in India, capable of providing round-the-clock service and spares support
- All these things put together, Cummins® offers you the

SINGLE SOURCE POWER ASSURANCE

Engine

- Cummins K19 series, 6 cylinder, In-line, 4 stroke, radiator cooled engine
- Highly stable and reliable design with square engine
- Well designed air handling system with
  - — Dry type, replaceable paper element air cleaner with restriction indicator
  - — Optimised turbocharger for increased altitude capabilities
  - — Air to air aftercooling
- Best in class fuel economy with
  - — PT fuel system with Electronic Step Timing Control (ESTC) injectors which smoothly stabilise engine speed under load with A1 class Electronic governing
  - — Dual fuel filter system: Pre filter including water separator and Water In Fuel (WIF) sensor and main filter
- Standard integral set-mounted radiator system, designed and tested for 50°C ambient temperature
- Spin on lube oil filter
- Plate type lube oil cooler
- First fill of lube oil and Coolant
- Electrical starter motor with soft start engagement feature
- Battery charging alternator
- 2 x 12 V DC Batteries

Alternator

- Stamford HC alternator frame from Cummins Generator Technologies
- Brushless Type, Screen protected, Revolving field, Self excited Alternator conforming to IS/IEC 60034-1
- 3 Phase reconnectable winding with 12 terminals brought out for connection
- Better motor starting capability
- Best in class efficiency
- Compact design with sealed bearings for longer life and lesser maintenance
- Impregnation on all wound components for better mechanical strength

Control Panel

Control panel is manufactured with 14/16 gauge CRCA sheet and is powder coated for weather-proof and long lasting finish. The control panel consists of the following parts:

- PowerCommand 1.2 Controller
- Aluminum bus bars with suitable capacity with incoming/ outgoing terminals
- Indicating lamps for 'Load ON' and 'Set Running'
- Instrument fuses duly wired and ferruled
- MCCB of suitable rating with overload and short circuit protections

PowerCommand 1.2 features

The PowerCommand® control system is a microprocessor-based generator set monitoring, metering and control system with LCD display designed to meet the demands of today's engine driven generator sets

- Intuitive operator interface which includes LED backlit LCD display with tactile feel soft-switches & generator set status LED lamps
- Digital AVR for shunt or PMG excitation with torque matching.
- Digital electronic governing with temperature compensation and smart starting.
- SAE J1939 interface to Full Authority Electronic (FAE) engines.
- Remote start-stop
- Engine metering: Oil pressure, Coolant temperature, Battery voltage, Engine speed
- AC Alternator metering: L-L Voltage and L-N Voltage, Current (1 and 3 phase), Volt-Amperes (phase and total) and Frequency.
- Engine protection: Low lube oil pressure, High/Low coolant temperature, Over speed, Battery Over/Under/Weak Volts, Fail to crank/start, Sensor failure.
- AC Alternator protection: Over/Under voltage, Over/Under frequency, Over current, Short circuit and Loss of AC sensing.
- Data logging: Engine hours, Control hours, Engine starts and upto 10 recent fault codes
- Configurable glow plug control
- Configurable cycle cranking
- 12 and 24 Volt DC operation
- Sleep mode
- Programmable I/Os (4 inputs and 2 outputs), expandable with AUX101/102 modules
- Modbus interface (RS485 RTU)
- InPower compatible (PC based service tool)
- Certifications - meets the requirement of relevant UL, NFPA, ISO, IEC, Mil Std., CE and CSA standards

Silencer

- Hospital grade silencer suitably optimised to meet stringent noise emission standards laid down by MoEF/CPCB

Mounting Arrangement

- Engine and alternator are mounted on a common MS fabricated base frame with AVM pads.
- Base frame with integral fuel tank is provided with drain plug, air vent, inlet and outlet connection, level indicator and provision for cleaning

Optional

- Engine: Coolant heater, Heavy duty air cleaner, Heat exchanger, No cool, Remote radiator
- Alternator: PMG
- Control Panel: PC3.3, Microprocessor / relay based AMF control panel
- Others: Trolley mounted mobile sets

Acoustic Enclosure

- Specially designed to meet stringent MoEF/ CPCB norms of 75 dBA @ 1mtr at 75% load under free field conditions
- The acoustic enclosure is made of CRCA sheets in Munsel green shade and a structural/ sheet metal base frame painted in black.
- High quality noise absorbant and fire-retardant grade acoustic insulation material (Rockwool) complying to IS 8183
- Base lifting for easy handling at customer site
- Designed to have optimum serviceability
- Air inlet louvers specially designed to operate at rated load
- Made on special purpose CNC machines for consistency in quality and workmanship
- 11 tank pretreatment process and UV resistant powder coating of all parts to withstand extreme
- environment
- Use of special hardware for longer life
- Flush styling - no projections
- Fluid drains for lube oil and fuel
- Fuel filling arrangement inside the enclosure

Technical Data

Generator Set Specification	
Model	C500D5P
Duty	Prime
Power Rating kVA / kWe	500/400
No. of Phases	3
Output Voltage and Frequency (V and Hz)	415 V, 50 Hz
Power Factor	0.8 (lagging)
Current (A)	695
RPM	1500
Engine Specification	
Make	Cummins
Model	KTAA19-G10
MoEF Certified Power (hp)	587
Required Power for Rated kVA (hp)	587
Cooling	Liquid Cooled (EG Compleat 50:50)
Aspiration	Turbocharged, Charge air Cooled
No. of cylinders	6, In-line
Bore(mm) x Stroke(mm)	159 x 159
Compression ratio	16.7:1
Displacement(litre)	19
Fuel	High Speed Diesel
Fuel consumption @75% load with radiator and fan*(litre/hr)	81.59
Fuel consumption @100% load with radiator and fan*(litre/hr)	106.28
Performance class of genset	ISO 8528-5 G2
Starting system	24 V DC Electrical
Lube oil specification	CH4 15W40
Lube oil sump capacity, High-Low level (litre)	38-32
Total lubrication system capacity (litre)	50
Lube oil consumption @ full load** (litre/hr)	0.15
Total coolant capacity (litre)	85
Exhaust pipe size (inch)	10
Total wet weight (Engine+Radiator) (Kg)##	2300
Length x Width x Height (Engine) (mm)	1744 x 916 x 1402
Mean piston speed (m/s)	7.95
Combustion air intake @100% load (±5%) (cfm)	1278
Exhaust Temperature (°C)	464
Alternator specification	
Make	Stamford (CGT)
Alternator frame	HCI544D
Enclosure	IP 23
Voltage regulation (Max.)	±1%
Class of Insulation	H Class
Winding Pitch	2/3
Stator Winding	Double layer lap
Rotor	Dynamically Balanced
Waveform distortion/ Total Harmonic Distortion	No load < 1.5 %, Non distorting balanced linear load < 5 %
Maximum Unbalanced Load across phases#	less than or equal to 25%
Telephonic Harmonic factor	< 2%

** Fuel consumption data is based on diesel having specific gravity of 0.85 and conforming to IS:1460. Fuel consumption tolerance is +5%*

***Oil consumption data is based on oil having specific gravity of 0.89 and meeting CH4 API categories*

 # With the condition that none of the phases exceeds its rated current

Rating Definitions

Prime Power (PRP):
Applicable for supplying power to varying electrical load for unlimited hours. Prime Power (PRP) is in accordance with ISO 8528

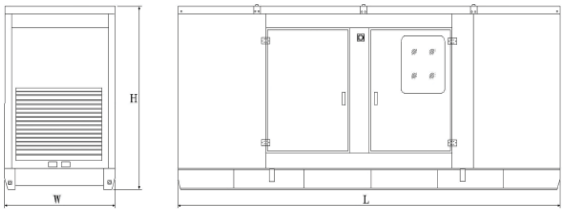
Conformance Standards

- IS/IEC 60034-1
- IS 1460
- ISO 8528
- ISO 3046
- ISO 9001
- IS 13018

Typical Enclosed Genset Dimensions

Genset Model	Rating (kVA)	Length (mm)	Width (mm)	Height (mm)	Wet Weight## (kg)	Standard Fuel tank Capacity (litre)
C500D5P	500	6000	2000	2250	7600	690

##Approximate weight



Authorised Representative

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Power Generation

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Power Generation

Production Test Certificate

General Information

WIP Job No:	13437902	Customer Order No:	UNASSIGNED
Customer Name:	UNASSIGNED	Test Cell No:	5

Product Information

Model:	C450 D6E	Genset Serial No:	J15K879646
Engine Serial No:	79865136	Alternator Serial No:	A15I399189
Fuel Type:	DIESEL	Controller Type:	PC2.2
Governor Type:	Electronic	Radiator Serial No:	N/A
Cooling System Type:	Mounted Radiator	Controller Serial No:	0065355D14

Test Conditions

Altitude (Meters ASL):	32	Barometric Pressure(mB):	10108
Ambient Temp(C):	14	Humidity(%):	27
Fuel Oil Spec:	BS 2869 Class A2	Lub Oil Spec:	Premium Blue 15W40
Fuel Density(kg/m):	820 - 875	Fuel Calorific Value:	43500 kJ/kg
Air Filter Inlet Temp(C):	14	Coolant Mix:	50/50 - Ethylene Glycol/Water

Product Options Checked

[X] Emergency Stop	Panel Heater	[X] Low Coolant S/Down	Low Fuel S/Down	High Fuel Warning
[X] Remote Start	Alternator Heater	Low Coolant Warning	Low Fuel Warning	Parallel Bus PT Module
Coolant Heater	Run Relay	[X] LET Warning	High Fuel S/Down	Earth Fault S/Down
CCB Trip Warning				

Technical Data - Declared

Service Rating:	STANDBY	Power Factor:	0.8
Power Output(kW):	450	Rated Output(kVA):	562
Rated Voltage(V):	380	Rated Current(A):	854
Frequency(Hz):	60	Speed(RPM):	1800
Number Of Phases:	3	Performance Class:	G2
Meter Serial No:	16071		

Technical Data - Measured

Results(No Load)

Voltage:	1-2:	383	Volts	2-3:	383	Volts	1-3:	383 Volts
	1-N:	221	Volts	2-N:	221	Volts	3-N:	221 Volts
No Load Freq:		60	Hz					
No Load Lube Press:		3.4	Bar				No Load Coolant Temp(C):	76

Results(Full Load)

Voltage:	1-2:	380	Volts	2-3:	380	Volts	1-3:	380 Volts
	1-N:	220	Volts	2-N:	220	Volts	3-N:	220 Volts
Current:	1:	683	Amps	2:	683	Amps	3:	683 Amps
Full Load Frequency:		60	Hz	Full Load Coolant Temp(C):	80		Power Factor:	1.0
Full Load Lube Press:		3	Bar	Prime set tested to 110% of Full Load(if r'eqd):				N/A

PTI/Static Test Completed By:	Pete Ayles	Date:	10-OCT-15		
Dynamic Test Completed By:	Norman Phillips	Date:	10-OCT-15	Run Hours:	0.55

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Technical File: Other Standards Applied

Machinery Directive, Directive 2006/42/EC: -	
SI 2008 No. 1597	The Supply of Machinery (Safety) Regulations 2008.
SI 2002 No. 1144	The Personal Protective Equipment Regulations 2002
BS EN 626-1:1994+A1:2008	Safety of machinery – Reduction of risks to health from hazardous substances emitted by machinery – Part 1: Principals and specifications for machinery manufacturers.
BS EN 626-2:1996+A1:2008	Safety of machinery – Reduction of risks to health from hazardous substances emitted by machinery – Part 2: Methodology leading to verification process.
BS EN ISO 3745:2012	Determination of sound power levels and sound energy levels of noise sources using sound pressure - Precision methods for anechoic rooms and hemi-anechoic rooms
BS EN ISO 3746:2010	Acoustics. Determination of sound power levels of noise sources using sound pressure. Survey method using an enveloping measurement surface over a reflecting plane
BS ISO 8528-1:2005	Reciprocating internal combustion engine driven alternating current generating sets. Application, ratings and performance.
BS ISO 8528-2:2005	Reciprocating internal combustion engine driven alternating current generating sets. Engines.
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets. Alternating current generators for generating sets.
BS ISO 8528-4:2005	Reciprocating internal combustion engine driven alternating current generating sets. Controlgear and switchgear.
BS ISO 8528-5:2005	Reciprocating internal combustion engine driven alternating current generating sets. Generating sets.
BS ISO 8528-6:2005	Reciprocating internal combustion engine driven alternating current generating sets. Test methods.
BS 7698-7:1996, ISO 8528-7:1994	Reciprocating internal combustion engine driven alternating current generating sets. Technical declarations for specification and design
BS 7698-8:1996, ISO 8528-8:1995	Reciprocating internal combustion engine driven alternating current generating sets. Requirements and tests for low-power generating sets
BS 7698-9:1996, ISO 8528-9:1995	Reciprocating internal combustion engine driven alternating current generating sets. Measurement and evaluation of mechanical vibrations
BS 7698-10:1999, ISO 8528-10:1998	Reciprocating internal combustion engine driven alternating current generating sets. Measurement of airborne noise by the enveloping surface method
BS 7698-12:1998, ISO 8528-12:1997	Reciprocating internal combustion engine driven alternating current generating sets. Emergency power supply to safety devices

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Transposed Harmonised Standards.

Machinery Directive 2006/42/EC.

Standard Number	Title
BS EN 1037:1995+A1:2008.	Safety of machinery – Prevention of unexpected start up.
BS EN 12601:2010.	Reciprocating internal combustion engine driven generating sets — Safety.
BS EN 349:1993+A1:2008.	Safety of machinery – Minimum gaps to avoid crushing parts of the human body.
BS EN 547-1:1996+A1:2008.	Safety of machinery. Human body measurements. Principles for determining the dimensions required for openings for whole body access into machinery.
BS EN 547-2:1996+A1:2008.	Safety of machinery. Human body measurements. Principles for determining the dimensions required for access openings.
BS EN 547-3:1996+A1:2008	Safety of machinery. Human body measurements. Anthropometric data.
BS EN 60204-1:2006+A1:2009.	Safety of machinery. Electrical equipment of machines. General requirements.
BS EN 614-1:2006+A1:2009.	Safety of machinery. Ergonomic design principles. Terminology and general principles.
BS EN ISO 13732-1:2008.	Ergonomics of the thermal environment. Methods for the assessment of human responses to contact with surfaces. Hot surfaces.
BS EN ISO 13850:2008.	Safety of machinery - Emergency stop. Principles for design.
BS EN ISO 13857:2008.	Safety of machinery. Safety distances to prevent hazard zones being reached by upper and lower limbs.
BS EN ISO 12100:2010	Safety of machinery. General principles for design. Risk assessment and risk reduction.
BS EN ISO 3744:2010.	Acoustics — Determination of sound power levels of noise sources using sound pressure — Engineering method in an essentially free field over a reflecting plane.

	essentially free field over a reflecting plane.
BS ISO 8528-1:2005	Reciprocating internal combustion engine driven alternating current generating sets. Application, ratings and performance.
BS 7698-10:1999, ISO 8528-10:1998	Reciprocating internal combustion engine driven alternating current generating sets. Measurement of airborne noise by the enveloping surface method.